# Shailza Jolly

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## Summary


- Builder and researcher with a Ph.D. in CS and 6+ years spanning academic research and industry, working at the intersection of science, engineering, and product. Specializes in LLMs, Generative AI, and scalable data pipelines, with hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design. Pairs a track record of top-tier publications (AAAI, NAACL, EMNLP) with shipping production systems and translating technical work into clear business outcomes.


## Experience

**Independent**, Applied AI & LLM Systems — Berlin, Germany — July 2025 – present — 1 year

- Exploring problem spaces in LLM evaluation, enterprise AI, and agentic systems by building working prototypes: Sift, a hybrid retrieval system for technical documentation, and a personal AI learning tool built after finding the gap in existing apps.
- Deepened LLM architecture fundamentals (Stanford) including efficiency internals: KV-cache, speculative decoding, FlashAttention.

**Flinn.ai**, Senior AI Engineer — Berlin, Germany — Jan 2025 – June 2025 — 6 months

- Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring.
- Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle.
- Navigated cost-performance tradeoffs under production constraints and learned to translate between engineers, product managers, and domain experts who each speak a fundamentally different language.

**Maternity**, Parental Leave — Berlin, Germany — Dec 2023 – Dec 2024 — 1 year 1 month

- Watched a human brain acquire language and reasoning from nothing; the most grounding perspective on what AI is actually trying to replicate.

**Amazon AI**, Research Scientist — Berlin, Germany — July 2022 – Nov 2023 — 1 year 5 months

- Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
- Presented findings and recommendations to senior science and engineering leadership; mentored Master's and Ph.D. interns on research deliverables.

**German Research Center for Artificial Intelligence (DFKI)**, Machine Learning Scientist — Berlin, Germany — Mar 2019 – June 2022 — 3 years 4 months

- Implemented transformer-based ML systems end-to-end (GPU training, Docker, production-level code) within BMBF-funded projects; published results at AAAI, NAACL, and EMNLP.
- Collaborated with academic and industry partners internationally; supervised interns and BSc/MSc students.

**NVIDIA Research**, Machine Learning Scientist Intern — Santa Clara, US — May 2021 – Aug 2021 — 4 months

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

**Amazon Alexa**, Applied Scientist Intern — Aachen, Germany — Aug 2019 – Dec 2019 — 5 months

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

**SAP AI Research**, Research Intern / Master's Thesis — Berlin, Germany
- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

May 2018 – Feb 2019
10 months

## Skills

**ML/GenAI:** LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation

**LLM Systems:** Agentic workflows (MCP, LangGraph/LangSmith), RAG, grounding, experiment design, offline evaluation

**Frameworks & Libraries:** Python, PyTorch, Hugging Face (Transformers, PEFT, Datasets), PySpark, Weights & Biases

**Search & Retrieval:** Sentence Transformers, reranking, vector search, Pinecone, Chroma

**Infrastructure & Deployment:** AWS, Docker, FastAPI, Git

## Education

**Ph.D.** **TU Kaiserslautern**, Computer Science — Kaiserslautern, Germany
- Grade: Sehr Gut 1.0 (highest on 1.0–5.0 scale)
- Thesis: Building Natural Language Generation and Understanding Systems in Data-Constrained Settings

2019 – 2022

**Visiting Researcher** **University of Copenhagen**, Natural Language Processing — Copenhagen, Denmark
- Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations

Jan 2021 – Mar 2021

**M.Sc.** **TU Kaiserslautern**, Computer Science — Minor in Economics — Kaiserslautern, Germany
- Grade: Sehr Gut 1.5

2016 – 2018

**Semester Abroad** **Kyushu University**, Computer Science — Fukuoka, Japan
- Explainable AI to analyze behavior of deep CNN architectures for image recognition

Nov 2017 – Feb 2018

**B.Tech** **Guru Nanak Dev Engineering College**, Computer Science & Engineering — India
- Grade: First Division with Distinction

2012 – 2016

## Selected Publications

**Search and Learn: Improving Semantic Coverage for Data-to-Text Generation** — Jan 2022
Shailza Jolly, Z. X. Zhang, A. Dengel, L. Mou
arxiv.org/abs/2112.02770 (AAAI)

**EaSe: A Diagnostic Tool for VQA Based on Answer Diversity** — Jan 2021
Shailza Jolly, S. Pezzelle, M. Nabi
www.aclweb.org/anthology/2021.naacl-main.192 (NAACL-HLT)

**Can Pre-training Help VQA with Lexical Variations?** — Jan 2020
Shailza Jolly, S. Kapoor
www.aclweb.org/anthology/2020.findings-emnlp.257 (Findings of EMNLP)

**Data-Efficient Paraphrase Generation to Bootstrap Intent Classification and Slot Labeling for New Features in Task-Oriented Dialog Systems** — Jan 2020
Shailza Jolly, T. Falke, C. Tirkaz, D. Sorokin
www.aclweb.org/anthology/2020.coling-industry.2 (COLING (Industry Track))

**How Do Convolutional Neural Networks Learn Design?** — Jan 2018
Shailza Jolly, B. K. Iwana, R. Kuroki, S. Uchida
ieeexplore.ieee.org/abstract/document/8545624 (International Conference on Pattern Recognition (ICPR))

## Awards

**AAAI-22 Scholarship by Hitachi** Jan 2022

**AI Newcomer 2021 Award** Jan 2021

- Recognized by the German Informatics Society and BMBF, Germany.

**EU-Cost STSM Grant** Jan 2020

- Awarded by EU-Cost Action to work on the Multi3generation project at CopeNLU, University of Copenhagen.

**Best Student Paper Award** Jan 2018

- Awarded by the International Conference on Pattern Recognition (ICPR) for "How Do Convolutional Neural Networks Learn Design?"

## Media Coverage

- Interview featured in the Rheinfalz newspaper.
- Radio interview for Antenne Kaiserslautern, Germany.

## Languages

**English:** Fluent

**German:** A2 (actively learning)

**Hindi:** Native speaker